

# Code Documentation:

## `dm_dephasing_full.py` / `dm_dephasing_colab.py`

Fisher-matrix distinguishability of dark-matter dephasing mechanisms in LISA observations

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### Abstract

This document describes, block by block, the Python implementation used to test whether a gravitational-wave dephasing signal detected by LISA can be *attributed* to a specific dark-matter environmental mechanism (gas/scalar-cloud accretion versus dynamical friction), or whether the two mechanisms are observationally degenerate. All dark-matter phase formulas are transcribed from Boudon, Brax, Valageas & Wong [1] with explicit equation numbers; the detector noise model is the full analytic LISA sensitivity of Robson, Cornish & Liu [2]; the Fisher-matrix formalism follows the stationary-phase-approximation treatment of Cutler & Flanagan [3] in the phase-only form used by [1].

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## 1 Overview and file structure

The code answers one question: given the joint Fisher information matrix over the binary parameters and *two simultaneous* dark-matter mechanism amplitudes, how close to unity is the off-diagonal correlation coefficient  $r_{AB}$  between the accretion density  $\rho_A$  and the dynamical-friction density  $\rho_B$ ? A value  $|r_{AB}| \rightarrow 1$  means the two mechanisms produce indistinguishable waveform-shape signatures over the observing band, so that *detection of a dephasing does not permit attribution of its physical origin*. The single-file Colab version (`dm_dephasing_colab.py`) contains the identical physics in the order documented below, followed by an analysis driver and figure generation.

The code is organised in eight blocks: (§2) physical constants, (§3) the LISA noise model, (§4) the binary-system class and regime thresholds, (§5) the frequency-domain phase model, (§6) the Fisher machinery, (§7) the per-system analysis, (§8) the parameter sweeps, and (§9) validation.

## 2 Block 1: physical constants

```
G = 6.67430e-11 # m^3 kg^-1 s^-2
c = 2.99792458e8 # m s^-1
Msun = 1.98892e30 # kg
GCM3 = 1.0e3 # 1 g/cm^3 in kg/m^3
YR = 3.15576e7 # s
FSTAR = 0.66 # critical-flux constant
```

All internal computations are in SI units; dark-matter densities are supplied and reported in  $\text{g cm}^{-3}$  (converted by `GCM3`), matching the units of the detection thresholds quoted in Table IV of [1]. The constant  $F_\star \simeq 0.66$  is the dimensionless critical accretion flux obtained numerically in [1] (Sec. II, around their Eq. (2.8)); it fixes the unique transonic radial-accretion solution of the self-interacting scalar cloud and enters both the low-frequency accretion phase and the accretion regime threshold.

**Implementation note.** An early version of this project used `G` both for Newton’s constant and for the Fisher matrix, which silently corrupted results; the Fisher matrix is therefore named `Gam/Gamma` throughout.

## 3 Block 2: LISA noise model (Robson, Cornish & Liu 2019)

Functions: `P_oms`, `P_acc`, `S_conf`, `Sn_LISA`. Source: [2], Eqs. (1), (10), (11), (14).

### 3.1 Single-link metrology and acceleration noise

`P_oms` implements the optical-metrology noise, Eq. (10) of [2],

$$P_{\text{OMS}}(f) = (1.5 \times 10^{-11} \text{ m})^2 \left[ 1 + \left( \frac{2 \text{ mHz}}{f} \right)^4 \right] \text{ Hz}^{-1}, \quad (1)$$

and `P_acc` the single test-mass acceleration noise, their Eq. (11),

$$P_{\text{acc}}(f) = (3 \times 10^{-15} \text{ m s}^{-2})^2 \left[ 1 + \left( \frac{0.4 \text{ mHz}}{f} \right)^2 \right] \left[ 1 + \left( \frac{f}{8 \text{ mHz}} \right)^4 \right] \text{ Hz}^{-1}. \quad (2)$$

### 3.2 Galactic confusion noise

`S_conf` is the analytic fit to the unresolved galactic-binary foreground, Eq. (14) of [2],

$$S_c(f) = A f^{-7/3} e^{-f^\alpha + \beta f \sin(\kappa f)} [1 + \tanh(\gamma(f_k - f))] \text{ Hz}^{-1}, \quad (3)$$

with the *4-year-mission* column of their Table 1:  $A = 9 \times 10^{-45}$ ,  $\alpha = 0.138$ ,  $\beta = -221$ ,  $\kappa = 521$ ,  $\gamma = 1680$ ,  $f_k = 1.13 \text{ mHz}$ . The 4-yr choice is consistent with  $T_{\text{obs}} = 4 \text{ yr}$  adopted by [1] (their Sec. VI.A).

### 3.3 Full sky-averaged sensitivity

`Sn_LISA` assembles Eq. (1) of [2],

$$S_n(f) = \frac{10}{3L^2} \left[ P_{\text{OMS}}(f) + \frac{4 P_{\text{acc}}(f)}{(2\pi f)^4} \right] \left[ 1 + \frac{6}{10} \left( \frac{f}{f_*} \right)^2 \right] + S_c(f), \quad (4)$$

with arm length  $L = 2.5 \times 10^9 \text{ m}$  and transfer frequency  $f_* = c/(2\pi L) = 19.09 \text{ mHz}$ . The prefactor  $10/3$  is the sky- and polarisation-averaged response penalty. Validation (§9): the code finds  $\min \sqrt{S_n} \simeq 1.18 \times 10^{-20} \text{ Hz}^{-1/2}$  at  $f \simeq 7 \text{ mHz}$ , matching the sensitivity floor of Fig. 3 in [2].

## 4 Block 3: the Binary class and regime thresholds

### 4.1 Mass variables

The constructor enforces the convention  $m_1 \geq m_2$  used throughout [1] and defines total mass  $m = m_1 + m_2$ , reduced mass  $\mu = m_1 m_2 / m$ , symmetric mass ratio  $\nu = \mu / m$ , and chirp mass

$$\mathcal{M} = \nu^{3/5} m, \quad (5)$$

which is Eq. (4.17) of [1].

### 4.2 Band edges

`lisa_band` implements the LISA row of Table I in [1]:

$$f_{\text{min}} = \max(2 \times 10^{-5} \text{ Hz}, f_{\text{obs}}), \quad f_{\text{max}} = \min(1 \text{ Hz}, f_\gamma). \quad (6)$$

Here `f_obs` is the GW frequency a time  $T_{\text{obs}}$  before merger (Sec. VI.A of [1], originally from their Ref. [113]),

$$f_{\text{obs}} = 4.149 \times 10^{-5} \left( \frac{\mathcal{M}}{10^6 M_\odot} \right)^{-5/8} \left( \frac{T_{\text{obs}}}{1 \text{ yr}} \right)^{-3/8} \text{ Hz}, \quad T_{\text{obs}} = 4 \text{ yr}, \quad (7)$$

and `f_gamma` is the relativistic-validity cut of Sec. IV.B.5 of [1]: the analysis stops where  $v^2 = 0.137 c^2$ , i.e., with the Keplerian relation  $v = (\pi G m f / c^3)^{1/3}$ ,

$$f_\gamma = 0.137^{3/2} \frac{c^3}{\pi G m}. \quad (8)$$

The ISCO frequency `f_isco`  $= c^3 / (6^{3/2} \pi G m)$  is provided for reference;  $f_\gamma < f_{\text{ISCO}}$ , so  $f_\gamma$  is the binding cut.

### 4.3 Accretion and dynamical-friction thresholds

The dark-matter phase contributions change functional form at characteristic frequencies controlled by the density ratio

$$y \equiv \frac{\rho_a}{\rho_0} = \frac{c^2}{c_s^2} \geq 1, \quad (9)$$

Eq. (5.4) of [1], where  $c_s$  is the effective sound speed of the self-interacting scalar cloud. The code inverts the threshold conditions, Eqs. (5.5)–(5.8) of [1], from bounds on  $y$  at fixed frequency into threshold frequencies at fixed  $y$ :

$$\text{f\_acc: accretion (Eqs. 5.5–5.6)} \quad f_{\text{acc},i} = \frac{c^3 m_i^3}{3\pi F_\star G m \mu^3 y}, \quad (10)$$

$$\text{f\_df\_minus (Eq. 5.8)} \quad f_{\text{df},i}^- = \frac{c^3 m_i^3}{\pi G m \mu^3 y^{3/2}}, \quad (11)$$

$$\text{f\_df\_plus (Eq. 5.7)} \quad f_{\text{df},i}^+ = \frac{e^3 c^3 m^5 \mu^{15}}{5832 \pi G m_i^{21} y^{3/2}}. \quad (12)$$

For  $f > f_{\text{acc},i}$  black hole  $i$  is in the high-frequency (velocity-regulated) accretion regime; dynamical friction on the smaller black hole operates in the window  $f_{\text{df},2}^- < f < f_{\text{df},2}^+$  (supersonic motion,  $v > c_s$ , with a nonempty Coulomb-logarithm range  $r_{\text{IR}} > r_{\text{UV}}$ ; see the discussion below Eq. (4.20) of [1]).

**Numerical-implementation note.** Equation (12) contains  $\mu^{15}$  and  $m_2^{21}$ ; in SI units ( $\mu \sim 10^{31}\text{--}10^{36}$  kg) these powers overflow IEEE double precision ( $\sim 10^{308}$ ). The methods `_log_fdf_plus_y1`, `f_df_plus` and `y_for_fdf_plus` therefore evaluate the expression in log space,  $\ln f_{\text{df}}^+ = 3 + 3 \ln c + 5 \ln m + 15 \ln \mu - \ln(5832 \pi G) - 21 \ln m_2 - \frac{3}{2} \ln y$ , and exponentiate at the end. The inverse map `y_for_fdf_plus` solves Eq. (12) for  $y$  given a target  $f_{\text{df}}^+$ ; it is used to *choose the fiducial self-interaction strength* such that the friction window brackets the observing band (§7).

## 5 Block 4: frequency-domain phase model

All phases are stationary-phase-approximation (SPA) Fourier phases  $\Psi(f)$  of the waveform  $\tilde{h}(f) = \mathcal{A}_0 f^{-7/6} e^{i\Psi(f)}$ , Eqs. (4.13)–(4.14) of [1] (the SPA construction itself follows [3]). The dimensionless frequency variable used everywhere is

$$x \equiv \frac{\pi G M f}{c^3}, \quad (13)$$

computed by `x_of_f`.

### 5.1 psi\_vacuum: vacuum GR phase, Eq. (4.16)

$$\Psi_{\text{gw}} = \frac{3}{128} x^{-5/3} \left[ 1 + \frac{20}{9} \left( \frac{743}{336} + \frac{11}{4} \nu \right) \left( \frac{\pi G m f}{c^3} \right)^{2/3} \right], \quad (14)$$

Eq. (4.16) of [1] (leading order plus 1PN; the PN coefficients originate from [3]). The 1PN term is essential: at leading order the phase depends only on  $\mathcal{M}$ , and the two masses would be exactly degenerate. Including 1PN breaks that degeneracy so the Fisher analysis over  $\{\ln m_1, \ln m_2\}$  is well defined — the same reason given in Sec. IV.B of [1]. Spin terms (1.5PN, 2PN) are omitted, as in [1].

## 5.2 `psi_halo`: cloud self-gravity, Eq. (4.18)

$$\Psi_{\text{halo}} = \frac{25\pi}{924} \frac{\rho_0 G^3 \mathcal{M}^2}{c^6} x^{-11/3}, \quad (15)$$

Eq. (4.18) of [1]. The exponent  $-11/3$  corresponds to an effective post-Newtonian order  $n = 3\alpha/2 + 5/2 = -3$  PN (their Sec. IV.C.4). The common dimensionless amplitude  $\rho_0 G^3 \mathcal{M}^2 / c^6$  is factored into the helper `_dm_prefac`, which also performs the  $\text{g cm}^{-3} \rightarrow \text{kg m}^{-3}$  conversion.

## 5.3 `psi_acc_high` / `psi_acc_low`: accretion, Eq. (4.19)

Equation (4.19) of [1] contains two branches separated by the thresholds (10).

**High-frequency branch** (`psi_acc_high`),  $f > f_{\text{acc},i}$ : proportional to the *bulk* density  $\rho_0$ ,

$$\Psi_{\text{acc}}^{(\text{high})} = -\frac{25\pi}{38912} \frac{G^3 \mathcal{M}^2 \rho_0}{c^6} x^{-16/3} \sum_i \Theta(f > f_{\text{acc},i}) \frac{m_i^3}{\mu^2 m} \left( 3 + \frac{2m_i^2}{m\mu} \right), \quad (16)$$

i.e., a  $-5.5$  PN contribution. The `include` argument implements the Heaviside factors  $\Theta$  at the fiducial: a black hole enters the sum only if its threshold lies below the band,  $f_{\text{acc},i} < f_{\text{min}}$  (cf. Eq. (5.10) of [1]). The mass structure  $\frac{m_i^3}{\mu^2 m} (3 + \frac{2m_i^2}{m\mu})$  is computed in `_acc_mass_factor`.

**Low-frequency branch** (`psi_acc_low`),  $f < f_{\text{acc},i}$ : proportional to the characteristic self-interaction density  $\rho_a$ ,

$$\Psi_{\text{acc}}^{(\text{low})} = -\frac{75\pi F_\star}{26624} \nu^{2/5} \frac{G^3 \mathcal{M}^2 \rho_a}{c^6} x^{-13/3} \sum_i \Theta(f < f_{\text{acc},i}) \left( 3 + \frac{2m_i^2}{m\mu} \right), \quad (17)$$

a  $-4$  PN contribution. The slowly varying bracket  $[1 - (f/f_{\text{acc},i})^{13/3} + \frac{13}{19}(f/f_{\text{acc},i})^{16/3}]$  of Eq. (4.19) is set to unity, which is the approximation stated explicitly below Eq. (5.10) of [1] (“the terms inside the brackets ... quickly converge to unity below the threshold”). In this project the low-frequency branch is used only as a comparison *shape* in the four-mechanism correlation matrix, not as a headline mechanism.

## 5.4 `psi_df`: dynamical friction, Eq. (4.20)

The main-window branch ( $f_{\text{df},2}^- < f < f_{\text{df},2}^+$ ) of Eq. (4.20) of [1], restricted to the smaller black hole (the paper’s leading  $\Theta$  factor: only  $m_2$  experiences significant friction):

$$\Psi_{\text{df}} = \frac{875\pi}{11829248} \frac{G^3 \mathcal{M}^2 \rho_0}{c^6} \frac{m_2^3}{\mu^2 m} x^{-16/3} \left[ 1 + \frac{304}{105} \ln \frac{f}{f_{\text{df},2}^+} - \frac{361}{105} \left( \frac{f}{f_{\text{df},2}^+} \right)^{16/3} + \frac{256}{105} \left( \frac{f}{f_{\text{df},2}^+} \right)^{19/3} \right]. \quad (18)$$

This is the physical crux of the whole analysis: dynamical friction carries the *same*  $x^{-16/3}$  ( $-5.5$  PN) carrier as high-frequency accretion, Eq. (16); the only shape difference is the slowly varying Coulomb-logarithm bracket (Sec. IV.C.4 of [1] assigns both mechanisms the same effective PN order). The power-law terms  $(f/f^+)^{16/3}$ ,  $(f/f^+)^{19/3}$  in the bracket are retained in the code; they matter only when  $f_{\text{df}}^+$  approaches the band. The second branch of Eq. (4.20)

( $f < f_{\text{df}}^-$ , where friction is degenerate with  $t_c$  and  $\Phi_c$ , their Eq. (5.11)) is not needed because the fiducial is chosen with the window covering the band.

**Linearity remark.** All four dark-matter phases are strictly linear in their density parameter at fixed  $y$  — the linearisation stated around Eq. (5.15) of [1] (“we treat  $\Psi_{\text{df}}$  as a linear function of  $\rho_0$  for a fixed density ratio  $y$ ”). The code exploits this: the Fisher derivative  $\partial\Psi/\partial\rho$  is simply the phase evaluated at  $\rho = 1 \text{ g cm}^{-3}$  (the “template”) — exact, with no finite-difference error.

## 6 Block 5: Fisher machinery

### 6.1 fisher: the phase-only Fisher matrix, Eq. (5.3)

For  $\tilde{h} = \mathcal{A}_0 f^{-7/6} e^{i\Psi}$  the amplitude  $\mathcal{A}_0$  decorrelates exactly from the phase parameters [3], and the general Fisher matrix

$$\Gamma_{ij} = 4 \text{Re} \int_{f_{\min}}^{f_{\max}} \frac{df}{S_n(f)} \partial_i \tilde{h}^* \partial_j \tilde{h} \quad (\text{Eq. (5.1) of [1]}) \quad (19)$$

reduces to the SNR-normalised, phase-only form implemented in `fisher`:

$$\Gamma_{ij} = \frac{(\text{SNR})^2}{\int_{f_{\min}}^{f_{\max}} \frac{df}{S_n(f)} f^{-7/3}} \int_{f_{\min}}^{f_{\max}} \frac{df}{S_n(f)} f^{-7/3} \frac{\partial\Psi}{\partial\theta_i} \frac{\partial\Psi}{\partial\theta_j}, \quad (20)$$

which is exactly Eq. (5.3) of [1]. The integrals are evaluated by trapezoidal quadrature on a 6000-point logarithmic frequency grid (`make_fgrid`). The SNR values are external inputs, taken from Table III of [1] (their Ref. [103]) for the benchmark events.

### 6.2 invert\_fisher: numerically stable inversion

Fisher matrices mixing parameters of vastly different scales ( $t_c$  in seconds vs.  $\rho$  in  $\text{g cm}^{-3}$  with  $\sigma \sim 10^{-22}$ ) are catastrophically ill-conditioned if inverted directly. The code first normalises to unit diagonal,

$$C_{ij} = \frac{\Gamma_{ij}}{\sqrt{\Gamma_{ii}\Gamma_{jj}}}, \quad \Sigma_{ij} = \frac{(C^{-1})_{ij}}{\sqrt{\Gamma_{ii}\Gamma_{jj}}}, \quad (21)$$

inverts the well-conditioned  $C$ , and rescales. This is a standard preconditioning identity (exact, not approximate). The routine returns the covariance  $\Sigma = \Gamma^{-1}$ , the standard deviations  $\sigma_i = \sqrt{\Sigma_{ii}}$  (cf. Sec. V.A of [1]), the correlation matrix

$$r_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}, \quad (22)$$

and the condition number of  $C$ , which is reported so the reader can verify the inversion is trustworthy.

For a  $2 \times 2$  sub-problem the exact identity

$$\frac{\sigma_{\text{joint}}}{\sigma_{\text{single}}} = \frac{1}{\sqrt{1 - r_{AB}^2}} \quad (23)$$

relates the correlation to the attribution penalty; the code verifies this numerically (e.g. for LISA-EMRI: degradation 63.9 vs.  $(1 - 0.999877^2)^{-1/2} = 63.8$ ), an internal consistency check of the whole Fisher pipeline.

### 6.3 binary\_param\_derivs: derivatives of the vacuum phase

The full phase is  $\Psi = 2\pi f t_c - \Phi_c - \pi/4 + \Psi_{\text{gw}} + \Psi_{\text{DM}}$ , Eq. (4.14) of [1], so

$$\frac{\partial \Psi}{\partial t_c} = 2\pi f, \quad \frac{\partial \Psi}{\partial \Phi_c} = -1 \quad (24)$$

are analytic. The mass derivatives  $\partial \Psi_{\text{gw}} / \partial \ln m_{1,2}$  are computed by central finite differences with relative step  $10^{-6}$  (rebuilding the `Binary` so that  $\mathcal{M}$ ,  $\nu$ ,  $m$  all respond consistently). The parameter set  $\{t_c, \Phi_c, \ln m_1, \ln m_2\}$  is exactly the binary set of Sec. V.A of [1].

## 7 Block 6: analyse\_system — the per-system analysis

This is the scientific core; it performs four Fisher analyses per system.

### 7.1 Fiducial self-interaction and regime bookkeeping

The friction-window edge  $f_{\text{df},2}^+$  sets how strongly the Coulomb-log bracket in Eq. (18) varies across the band. The code places it at  $f_{\text{df},2}^+ = R f_{\text{max}}$  (default  $R = 10$ ) and inverts Eq. (12) for the corresponding density ratio  $y$  (equivalently  $c_s = c/\sqrt{y}$ ). It then computes  $f_{\text{acc},1}$ ,  $f_{\text{acc},2}$ ,  $f_{\text{df},2}^-$  and records which regime assumptions hold: (i) the friction window covers the band, (ii) which black holes are in the high-frequency accretion regime (these enter `acc_include`), and (iii) whether friction is active at all — for comparable masses ( $\nu > 0.16$ , mass ratio  $\lesssim 2.6$ ) the window closes and there is no dynamical friction, as noted for the MBBH case in Sec. VI.C of [1]. All thresholds are printed so every run documents its own validity.

### 7.2 Analysis (a): single-mechanism detection thresholds

For each mechanism template  $t \in \{\text{accretion, friction, halo, low-}f \text{ accretion}\}$  a  $5 \times 5$  Fisher matrix over  $\{t_c, \Phi_c, \ln m_1, \ln m_2, \rho\}$  is built and inverted; the reported  $\sigma(\rho_0)$  is the  $1\sigma$  detection threshold  $\rho_{0*} = \sigma_{\rho_0}$ , the same criterion as Sec. V.B.2 of [1]. This reproduces the *detectability* analysis of the literature and provides the baseline against which attribution is compared.

### 7.3 Analysis (b): the joint accretion+friction Fisher matrix

The  $6 \times 6$  matrix over  $\{t_c, \Phi_c, \ln m_1, \ln m_2, \rho_A, \rho_B\}$  with *both* mechanisms present simultaneously. Its inverse yields the headline quantities: the marginalised correlation coefficient  $r_{AB} = \Sigma_{\rho_A \rho_B} / (\sigma_{\rho_A} \sigma_{\rho_B})$  and the degradation factors  $\sigma_{\text{joint}} / \sigma_{\text{single}}$  (Eq. (23)). This joint construction is the project’s novel step: [1] always analyse one environment model at a time, so the question of *mechanism attribution* — whether the data can tell which physics produced a detected dephasing — is not addressed there.

### 7.4 Analysis (c): DM-only correlation

The same two templates without the binary parameters ( $2 \times 2$  Fisher). Comparing  $r_{AB}^{(\text{DM only})}$  with the marginalised  $r_{AB}$  isolates how much of the confusion is intrinsic to the two dark-matter shapes versus induced by marginalising over  $\{t_c, \Phi_c, \ln m_1, \ln m_2\}$ .

### 7.5 Analysis (d): four-mechanism correlation matrix

An  $8 \times 8$  Fisher over the binary parameters plus all four mechanism amplitudes, reported as the marginalised  $4 \times 4$  mechanism-correlation block. This generalises the pairwise question: it shows that after marginalisation even mechanisms separated by 1.5–2.5 effective PN orders (−3 PN halo vs. −5.5 PN accretion) correlate at  $|r| > 0.99$  over a one-decade band.

## 8 Block 7: parameter sweeps

### 8.1 sweep\_mass

Repeats the full analysis for  $m_1 \in [10^4, 10^7] M_\odot$  at fixed mass ratio  $q = 10^4$  and fixed SNR = 30 (so that the sweep isolates the *shape* degeneracy from signal loudness;  $r_{AB}$  is SNR-independent in the Fisher formalism, while thresholds scale as  $1/\text{SNR}$ ). The band is recomputed per system from `lisa_band`. Output:  $1 - |r_{AB}|$  and the degradation factor vs.  $m_1$  (Fig. 3 of the results).

### 8.2 sweep\_fdfplus

Repeats the analysis varying the friction-window placement  $R = f_{\text{df},2}^+ / f_{\text{max}} \in [1.1, 1000]$ , which through Eq. (12) is a scan over the self-interaction ratio  $y$  (equivalently  $c_s$ ). Physical meaning: when  $R \rightarrow 1$  the logarithmic turnover of Eq. (18) sits inside the band and the friction template is maximally distinct from the accretion power law; as  $R \rightarrow \infty$  the bracket tends to a constant and the two templates become identical,  $|r_{AB}| \rightarrow 1$ . This sweep demonstrates that the degeneracy is not an artefact of one fiducial choice (Fig. 5 of the results).

## 9 Block 8: validate — checks against the sources

Run before any result is trusted; each check targets a specific published number.

| Check                                                                 | Code result                         | Reference                   | Source          |
|-----------------------------------------------------------------------|-------------------------------------|-----------------------------|-----------------|
| Noise floor $\min \sqrt{S_n}$                                         | $1.18 \times 10^{-20}$ at 7.3 mHz   | sky-averaged floor          | [2], Fig. 3     |
| $\Psi_{\text{halo}}/\Psi_{\text{gw}}$ at 1 Hz, $\rho_0 = 1$           | $2.45 \times 10^{-8}$ , mass-indep. | $\sim 2 \times 10^{-8}$     | [1], Eq. (4.21) |
| $ \Psi_{\text{acc},>} /\Psi_{\text{gw}}$ , $q=2$                      | 1.8                                 | $\sim 0.5$ (order-of-magn.) | [1], Eq. (4.23) |
| MBBH $\sigma(\rho_0)$ , SNR $3 \times 10^4$                           | $6.2 \times 10^{-12}$               | $\sim 8 \times 10^{-13}$    | [1], Table IV   |
| $\sigma_{\text{joint}}/\sigma_{\text{single}}$ vs. $(1 - r^2)^{-1/2}$ | 63.9 vs. 63.8                       | exact identity              | Eq. (23)        |

Two of these deserve comment. Equation (4.21) of [1] predicts a mass-independent halo-to-vacuum phase ratio; the code reproduces both the value and the mass independence, a strong test of the Eq. (4.18) prefactor. The MBBH threshold agrees only in order of magnitude: since  $\sigma(\rho_0) \propto f_{\text{min}}^{16/3}$  (Eq. (5.14) of [1]), a factor  $\sim 1.5$  difference in the low-frequency cutoff between the noise model used here ([2]) and theirs fully accounts for the factor  $\sim 8$ ; exact agreement is not expected and this sensitivity should be stated in the paper’s methods section.

## 10 Analysis driver and figure code (Colab file)

`run_analysis()` executes: validation, the two friction-active LISA benchmarks of Table II/III of [1] (IMRI  $10^4 + 10 M_\odot$ , SNR 64; EMRI  $10^5 + 10 M_\odot$ , SNR 22 — the MBBH is excluded from the joint analysis because  $\nu > 0.16$  switches friction off), both sweeps, and the CSV outputs (`results_systems.csv`, `sweep_mass.csv`, `sweep_fdfplus.csv`).

`make_all_figures()` renders five figures with the non-interactive Agg backend (files on disk rather than inline display — in a notebook, view them with `IPython.display.Image`): Fig. 1, the sensitivity (4) with the per-system bands; Fig. 2, the four normalised templates and the friction/accretion shape ratio (making the shared  $-16/3$  carrier visible); Fig. 3,  $1 - |r_{AB}|$  vs.  $m_1$  with the degradation factor; Fig. 4, the collapsed joint  $(\rho_A, \rho_B)$  error ellipse (eigen-decomposition of the marginalised  $2 \times 2$  covariance block); Fig. 5,  $1 - |r_{AB}|$  vs.  $f_{\text{df}}^+ / f_{\text{max}}$ .



## References

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